ZUU CREW · DWDF CAPSTONE

PulseTrack Dataset

Data Guide — a raw, unnormalised, THREE-source performance-marketing platform

Relational export (11 tables) + MongoDB (2 collections) + Kafka-style event stream · ~958,000 records · one seeded ground truth — the sources reconcile exactly

For the instructor: this package is the DATA only (schemas + ER + this guide + the raw files + answer key). Build the project brief around it.

Version 2.0 · 2026

1. What this dataset is

PulseTrack is a fictional performance-marketing / affiliate-tracking platform (think of a Cake or Everflow). Version 2 delivers its data the way a real client would: through THREE different systems, each with its own format and its own problems:

- **1. The relational export** (data/*.csv + ddl/) — 11 related tables, unnormalised, redundant and dirty. The platform's nightly database dump.
- **2. The PartnerHub / CRM in MongoDB** (mongo/*.jsonl) — nested JSON documents: partner portal profiles and the advertiser CRM, including the account-manager history.
- **3. The conversion event stream** (stream/pulsetrack_conversions.log) — a Kafka-style topic dump: the real-time event view of the same conversions the batch export shows as state.

All three are generated from ONE seeded ground truth, so they genuinely reconcile: students can (and must) combine them while normalising — resolving identities across sources, choosing precedence when sources conflict, and proving that the replayed stream matches the batch state. The cleaning, conforming and reconciling ARE the work.

The business, in one paragraph

Advertisers (brands) run campaigns that promote offers, using creatives (ads). Affiliate partners drive traffic: impressions → clicks → conversions. Each conversion pays the partner a payout and earns the advertiser revenue; revenue is edited later by adjustments. A conversion can be credited to several clicks (multi-touch). Partners self-manage in a portal (MongoDB); advertisers are managed in a CRM (MongoDB); every conversion fires real-time events (the stream).

2. The three sources at a glance

Source	Format	Contents	Size
Relational export	11 CSVs + PostgreSQL DDL	the full funnel + reference tables (unnormalised, dirty)	774,344 rows · ~99 MB
MongoDB PartnerHub/CRM	2 JSONL collections (mongoexport style)	partner_profiles (600 docs) · advertiser_accounts (60 docs)	660 docs · ~0.3 MB
Event stream	Kafka-style topic dump (1 line = 1 record)	conversion.created / status_changed / adjustment.applied	183,162 lines · ~63 MB

New in v2: every relational table carries two audit columns — `created_at` and `updated_at` (messy formats; ~1–3% of rows have updated_at BEFORE created_at — a validation catch; ~5% null) — EXCEPT on conversions_raw, where updated_at is the dedup key: never null, never inverted, always time-bearing. They enable watermark-based incremental extraction. Timezone convention: every wall-clock time in the dataset is UTC; epoch values are true UTC epochs.

The 11 relational tables

Table	≈ Rows	Role	What it primarily teaches
account_managers_raw	25	reference	the SCD2 subject (AM ownership)
advertisers_raw	60	reference	name canonicalisation; current AM
partners_raw	600	reference	identity resolution (id prefix)
offers_raw	150	reference	embedded-name reconciliation
campaigns_raw	400	reference	the snowflake chain + orphans
creatives_raw	1,200	reference	deepest dimension in the chain
impressions_raw	300,000	factless fact	funnel top; junk columns; volume
clicks_raw	150,000	event fact	junk/PII; mixed booleans; funnel
conversions_raw	126,380 (90,000 ids)	transactional fact	the dirty, unnormalised centrepiece; dedup by MAX(updated_at)
conversion_touchpoints_raw	183,529	bridge (M:N)	multi-touch attribution
revenue_adjustments_raw	12,000	late-arriving	incremental / idempotent + true revenue

3. Source relationships (ER)

The reference tables form a snowflake chain (creatives → campaigns → offers → advertisers → account_managers); the events form a funnel (impressions → clicks → conversions); touchpoints bridge conversions to their clicks (M:N); adjustments hang off conversions. conversions_raw ALSO embeds the advertiser / campaign / partner / offer names redundantly — reconciling those against the reference tables is the normalisation exercise. The bottom band shows the two companion sources and the cross-source keys.

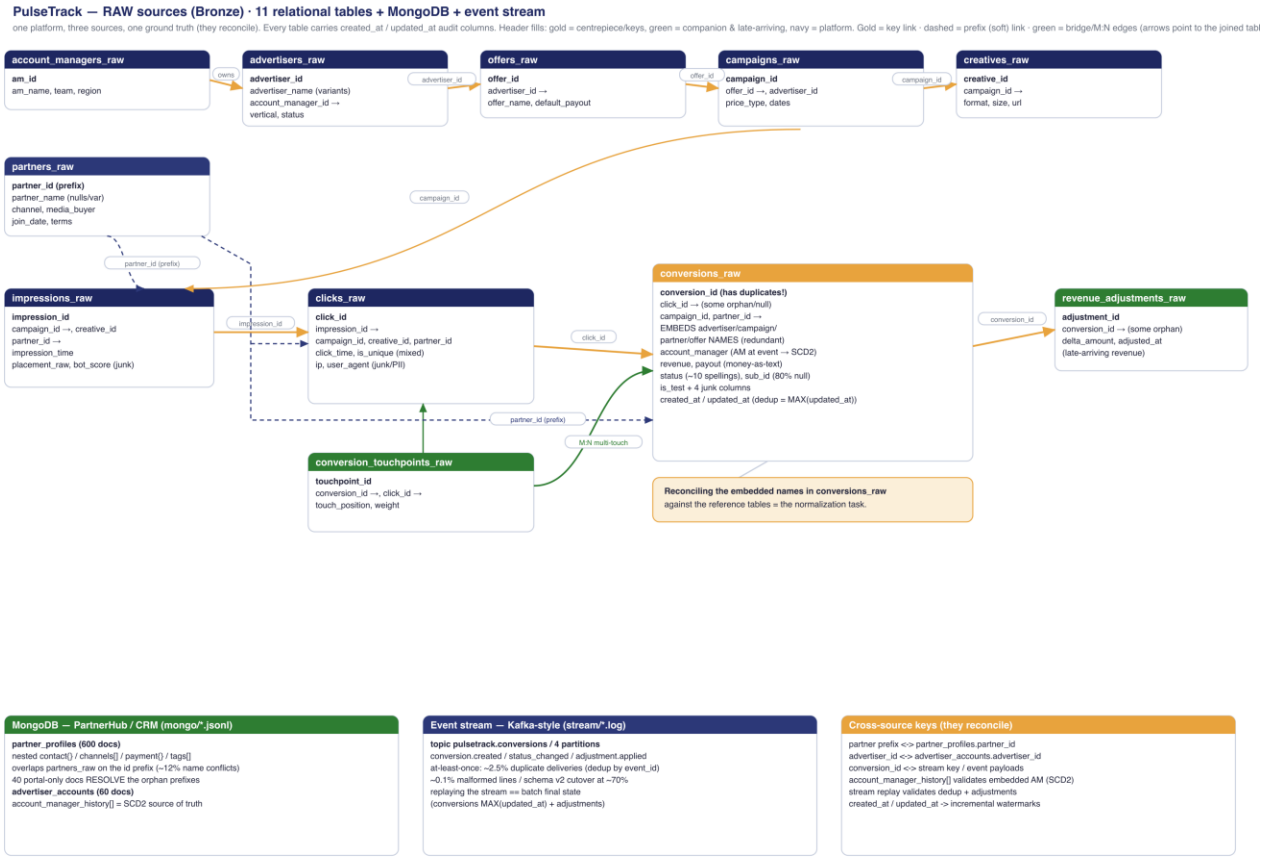


Figure 1. PulseTrack raw sources — 11 relational tables + the two companion sources. (Full-resolution PNG: pulsetrack_er_diagram.png)

4. Student quickstart — how to attack this dataset

A proven order of attack (each step builds on the last):

- **1. Profile first.** Load the 11 CSVs as ALL-TEXT (Bronze). Count rows, nulls and distinct values per column before cleaning anything.
- **2. Clean into Silver.** Parse money and dates with the cheat-sheet below; standardise status / price_type / country / device to clean enums; filter is_test rows.
- **3. De-duplicate conversions.** Keep the MAX(updated_at) row per conversion_id — updated_at is never null there and always carries a time.
- **4. Resolve identities.** Extract the 5-digit partner prefix; union partners_raw with the MongoDB partner_profiles (the 40 unknown prefixes resolve there); decide a name-precedence rule.
- **5. Model the star.** Conformed dimensions + fact_conversion at the deduplicated grain; build the SCD2 advertiser dimension from account_manager_history[] in MongoDB.
- **6. Replay the stream.** Dedup by event_id, fold events per conversion — your replayed final state must equal your batch dedup, and stream adjustment deltas complete net revenue (batch delta_amount is sometimes N/A).
- **7. Reconcile & report.** Prove batch == stream, quantify what you quarantined (malformed lines, invalid rows), then build your marts/dashboards on Gold.

Parsing cheat-sheet

Field family	Formats you will meet	Parse hint
Money	$1,234.56 · USD 12.34 · 12.34 USD · 1234.56 · $ 1,234.56 · N/A · blank · negatives	strip $, USD, US$, commas; N/A and blank are NULL; keep negatives (real anomalies)
Dates (general)	YYYY-MM-DD [HH:MM:SS] · MM/DD/YYYY · Mon DD, YYYY · DD-MM-YYYY · epoch seconds · ISO with tz suffix	SLASH dates are MM/DD/YYYY; DASH dates are DD-MM-YYYY; some suffixes (‘ EST’, offsets) are noise
updated_at (conversions)	YYYY-MM-DD HH:MM:SS · epoch seconds · ISO…Z · MM/DD/YYYY HH:MM	always time-bearing, never null — safe as the dedup / watermark key
Timezone convention	every wall-clock time is UTC; epochs are true UTC epochs	set your session timezone to UTC (e.g. DuckDB: SET TimeZone='UTC') or naive-vs-epoch comparisons silently shift
status	approved/Approved/APPROVED/app/1 · pending/Pending · reversed/Reversed/returned	canonicalise to approved / pending / reversed
partner_id	12345_678901 (compound) · 12345 (bare) · plat<hex> · blank	canonical key = the 5-digit prefix; plat/blank are unattributable
Booleans	1/0 · true/false · Y/N · blank	normalise; blank = unknown
is_test	test · 1 · true · 0 · blank	filter only the TRUTHY values

Known limitations (by design — not your bugs)

- Row-level funnel chronology is NOT guaranteed: a click's timestamp may precede its linked impression's. The dataset teaches set-level joins and aggregation, not sessionization.
- The first and last calendar months are thin — late lifecycle updates spill past the export window. Analyse the core months (2025-02 to 2026-05).
- Some batch money values are N/A/blank on purpose: the STREAM is the only complete source — exact net revenue requires folding stream events.
- The batch export holds every conversion's FINAL state but not every intermediate state; the stream holds the full history. Transition counts differ between sources — that is the lesson, not an error.

5. Relational source — table-by-table

> **Note:** *Every table below ALSO carries created_at / updated_at (audit columns; mixed formats; ~1–3% updated_at earlier than created_at; ~5% null — except conversions_raw.updated_at, the dedup key: never null, never inverted). They are omitted from the per-table lists to avoid repetition.*

5.1 account_managers_raw

Internal account managers. Mostly clean — the SCD2 subject (an advertiser's owning AM changes over time).

Column	Notes / deliberate issue
am_id	key (e.g. AM007)
am_name	descriptive — exactly one null (AM013, deliberate)
team, region	attributes
hire_date	mixed date formats

5.2 advertisers_raw

Brands / clients. Shows the CURRENT account manager only — the full history lives in MongoDB (advertiser_accounts.account_manager_history).

Column	Notes / deliberate issue
advertiser_id	key (ADV0001)
advertiser_name	casing variants
vertical, status	attributes
account_manager_id	→ account_managers_raw (current owner)
signup_date	messy dates
billing_country	attribute

5.3 partners_raw

Affiliates / publishers. The CANONICAL partner id is the numeric prefix. 40 partners that appear in events are MISSING here — they exist only in MongoDB partner_profiles (cross-source resolution).

Column	Notes / deliberate issue
partner_id	key = numeric prefix (e.g. 42266)
partner_name	~12% null / variants
channel	casing variants
media_buyer, payment_terms	attributes
join_date	messy dates

5.4 offers_raw

Offers / products promoted (offer_name is also embedded in conversions).

Column	Notes / deliberate issue
offer_id	key (OFF0001)
advertiser_id	→ advertisers_raw
offer_name	descriptive
default_payout	money-as-text
vertical, payout_type	attributes

5.5 campaigns_raw

Campaigns promoting an offer.

Column	Notes / deliberate issue
campaign_id	key (CMP00001)

Column	Notes / deliberate issue
offer_id	→ offers_raw
advertiser_id	→ advertisers_raw (~3% — 12 campaigns with invalid ids that contradict the offer's advertiser)
campaign_name	casing variants
price_type	CPA/CPC/CPM/Fixed + casing noise
start_date, end_date	messy dates

5.6 creatives_raw

Ads within a campaign (deepest node in the snowflake chain).

Column	Notes / deliberate issue
creative_id	key (CRE00001)
campaign_id	→ campaigns_raw
creative_name, format, size, landing_url	attributes

5.7 impressions_raw

FACTLESS fact — funnel top (high volume).

Column	Notes / deliberate issue
impression_id	key
campaign_id, creative_id, partner_id	→ reference tables (creative_id blank ~5%: campaign without creatives)
impression_time	mixed formats + timezones
placement_raw, bot_score	JUNK columns to ignore
country, device	inconsistent categoricals

5.8 clicks_raw

Click events (funnel middle). Most clicks never convert.

Column	Notes / deliberate issue
click_id	key
impression_id	→ impressions_raw (NULL for ~15% 'direct')
campaign_id, creative_id, partner_id	references
click time	mixed formats + timezones
is_unique	mixed booleans (1/0/true/false/Y/N/blank)
ip, user_agent	JUNK / PII (governance/tagging hook)
country, device	inconsistent categoricals
referrer_raw	JUNK (nullable)

5.9 conversions_raw

TRANSACTIONAL fact and the UNNORMALISED, DIRTY centrepiece. conversion_id is NOT unique: a conversion MAY appear multiple times — up to once per lifecycle state, with increasing updated_at. The FINAL state is ALWAYS present; the nightly export did not capture every intermediate state (the stream holds the full history). Plus ~2% exact duplicates. THE RULE: the correct row is MAX(updated_at) per conversion_id.

Column	Notes / deliberate issue
conversion_id	key — 126,380 rows / 90,000 distinct ids
click id	→ clicks raw (some orphan / null)
campaign_id, partner_id	references (partner_id compound/null)

Column	Notes / deliberate issue
advertiser_name / campaign_name / partner_name / offer_name	EMBEDDED, redundant, inconsistent — reconcile to dims
account_manager	AM valid AT event time → SCD2 signal (validate against MongoDB am_history)
price_type	casing noise (CPA/CPC/CPM/Fixed variants)
conversion_time	mixed date formats + timezones
revenue, payout	money-as-text ($ / USD / plain / N/A / blank / negative)
currency	USD / usd / $ / US$ / blank
status	~10 spellings/codes for approved / pending / reversed
country, device	inconsistent categoricals
sub_id	~80% null
is_test	mixed flags (test / 1 / true / 0 / blank) — filter truthy rows out
legacy_flag, pixel_fired, internal_note, reserved_1	JUNK columns (internal_note is entirely empty)
updated_at	THE DEDUP KEY — never null; formats: 'YYYY-MM-DD HH:MM:SS', epoch seconds (UTC), ISO…Z, 'MM/DD/YYYY HH:MM'; keep MAX per id

5.10 conversion_touchpoints_raw

BRIDGE (many-to-many): many clicks → one conversion (multi-touch attribution).

Column	Notes / deliberate issue
touchpoint_id	key
conversion_id	→ conversions_raw
click_id	→ clicks_raw
touch_position	first / mid / last
weight	fractional attribution weight (sums to ≈1 per conversion — 3-decimal rounding)

5.11 revenue_adjustments_raw

LATE-ARRIVING revenue edits applied after the conversion. Each row also appears as an adjustment.applied event in the stream. NOTE: ~8% of batch delta_amount values are blank/'N/A' (and some winning revenue strings are 'N/A') — the STREAM is the only complete source; exact net revenue requires folding stream adjustment events.

Column	Notes / deliberate issue
adjustment_id	key
conversion_id	→ conversions_raw (~6% ORPHANED — conversions outside the export window)
delta_amount	money-as-text (may be NEGATIVE; ~8% blank/'N/A' — recover the value from the stream)
reason	free text
adjusted_at	messy date, later than the conversion

6. Companion source — MongoDB (PartnerHub / CRM)

Two collections, shipped as mongoexport-style JSONL (one document per line, with $oid / $date extended-JSON). Load them into a real MongoDB (mongoimport) or parse the JSONL directly — both are legitimate paths.

6.1 partner_profiles (600 docs)

- **Shape:** `{_id, partner_id, name, contact{person,email,phone}, channels[], payment{terms,min_payout,method}, tags[], joinedAt, createdAt, updatedAt}`
- **Overlap:** 560 docs share the numeric prefix with partners_raw; ~12% of those carry a CONFLICTING name (the portal-registered variant, e.g. 'Reach Labs Group' vs 'Reach Labs'). Students must define a precedence rule.
- **The payoff:** 40 docs are PORTAL-ONLY — they are exactly the 40 orphan prefixes that appear in events but are missing from partners_raw. Combining the two sources resolves the orphans.
- **Document mess:** ~10% of docs missing the tags field, ~5% carry an extra beta_features field (schema drift); dates are a mix of {$date: ISO} and plain strings; min_payout is sometimes a number, sometimes a string (mixed types); nested nulls in contact.

6.2 advertiser_accounts (60 docs)

- **Shape:** `{_id, advertiser_id, legal_name, display_name, vertical, status, billing{}, contacts[], account_manager_history[], contract{payout_tiers[]}, createdAt, updatedAt}`
- **account_manager_history[] is the SCD2 SOURCE OF TRUTH:** each entry has am_id, am_name, from, to (null = current). It reconciles exactly with the account_manager embedded in conversions_raw at event time — students build the SCD2 dimension from this array and validate it against the embedded signal.
- **Arrays to unnest:** contacts[], payout_tiers[], account_manager_history[] — document-to-relational flattening practice. legal_name vs display_name vs the relational advertiser_name = another canonicalisation decision.

```
# load into MongoDB (optional — parsing the JSONL directly is equally valid)
mongoimport --db pulsetrack --collection partner_profiles   --file mongo/partner_profiles.jsonl
mongoimport --db pulsetrack --collection advertiser_accounts --file mongo/advertiser_accounts.jsonl
```

7. Companion source — the event stream (Kafka-style)

stream/pulsetrack_conversions.log is a dump of the topic pulsetrack.conversions exactly as a consumer of all 4 partitions would see it — one record per line:

```
pulsetrack.conversions|partition=2|offset=18452|ts=1781470801000|key=CV0012345|{"event_id":"EV-CV0012345-
1","event_type":"conversion.status_changed","conversion_id":"CV0012345","old_status":"pending","new_
status":"approved","occurred_at":"2026-06-14T21:00:01Z"}
```

Property	Value / rule
Volume	183,162 lines · 4 partitions · per-partition offsets strictly increasing
Event types	unique events after event_id dedup: conversion.created (90,000) · conversion.status_changed (one per lifecycle transition) · adjustment.applied (12,000 — mirrors revenue_adjustments_raw)
At-least-once delivery	~2.5% duplicate deliveries: the SAME event_id re-emitted at a later offset. De-duplicate by event_id.
Malformed lines	184 truncated lines (~0.1%) — parse defensively, quarantine and count them.
Schema evolution	after ~70% of the stream (the 125,019th unique event, 0-indexed) payloads carry schema_version '2.0' and conversion.created renames revenue → revenue_amount.
Ordering	guaranteed only WITHIN a partition — a per-conversion timeline must be rebuilt from occurred_at / offsets.
Envelope ts	producer event-time in epoch millis (UTC). Duplicate deliveries carry a slightly later ts; offsets are the authoritative order.
Test traffic	is_test conversions also emit events — filter them, as in the batch source.

The reconciliation exercise (the reason this stream exists): replay the stream — dedup by event_id, fold events per conversion_id, take the last status — and the result MUST equal the batch state: conversions_raw deduplicated by MAX(updated_at), with net revenue = revenue + the summed adjustment deltas. Event-sourced view vs state view of the same truth.

8. Cross-source keys & how the sources combine

Key	Connects	Use during ETL
partner prefix	partners_raw ↔ partner_profiles.partner_id ↔ events' compound partner_id	one conformed dim_partner; Mongo fills the 40 relational orphans; precedence rule for name conflicts
advertiser_id	advertisers_raw ↔ advertiser_accounts.advertiser_id ↔ campaigns/offers	one conformed dim_advertiser; CRM legal/display names vs relational name
am_history[] (Mongo)	account_managers_raw ↔ the AM embedded in conversions	build the SCD2 dimension; assign each conversion its historically-correct AM key
conversion_id	conversions_raw ↔ stream key/payloads ↔ touchpoints ↔ adjustments	stream replay validates batch dedup; adjustments appear in BOTH stream and batch (must match 1:1)
created_at / updated_at	every relational table	incremental extraction watermarks; conversions dedup (MAX(updated_at))

> **Note:** *Everything reconciles because all three sources are emitted from one seeded ground truth. Every deliberate exception (orphans, conflicts, anomalies) is catalogued in §9 — there are no accidental mismatches.*

9. Data-issue catalog (the full cleaning surface)

Data issue	Where it lives	What it teaches
Money as text ($ / USD / plain / N/A / blank / negative)	revenue, payout, default_payout, delta_amount	parse & validate monetary values
Date + timezone chaos (ISO / US / epoch / tz-suffixed)	all *_time / *_date / audit columns	date parsing & standardisation
Casing & name variants (LaserAway / laseraway / LaserAway, LLC)	advertiser/campaign/partner/channel names	canonicalisation → conformed dimensions
Compound / bare / plat / null partner_id	conversions, clicks, impressions	identity resolution (prefix-keying)
Redundant EMBEDDED names (unnormalised)	conversions_raw	normalisation into dimensions
Duplicate rows — lifecycle snapshots + exact dups	conversions_raw (126,380 rows / 90,000 ids)	de-duplication: keep MAX(updated_at)
Missing intermediate snapshots	conversions_raw holds every FINAL state but not every intermediate one; the stream has the full history	event-vs-state thinking; why streams exist
Orphaned foreign keys	click_id, advertiser_id, conversion_id, 40 partner prefixes	referential integrity; the partner orphans are RESOLVED via MongoDB
Late-arriving revenue	revenue_adjustments_raw + adjustment.applied events	incremental / idempotent + true revenue
Audit-column anomalies	~1–3% updated_at < created_at; ~5% null (never on conversions_raw.updated_at — the dedup key)	validation gates; watermark edge cases
Null-heavy columns	sub_id (~80%), partner_name (~12%)	null handling / data-quality gates
Out-of-range values	negative revenue (~3%), payout > revenue (~5%) — consistent across batch AND stream	validation / quarantine
Inconsistent categoricals	status, price_type, country, device	standardise to clean enums
Ambiguous date-only formats	slash dates = MM/DD/YYYY; dash dates = DD-MM-YYYY; some time columns carry a bogus ' EST' suffix	explicit format contracts
Junk / dummy columns	placement_raw, bot_score, legacy_flag, pixel_fired, internal_note, reserved_1, referrer_raw	relevance-spotting (drop them)
Attribute change over time	Mongo am_history[] + embedded AM in conversions	Slowly Changing Dimension (Type 2)
Many clicks → one conversion	conversion_touchpoints_raw	bridge table / M:N modelling
Document schema drift	partner_profiles: missing/extra fields, mixed types, $date vs string	semi-structured ingestion
Cross-source name conflicts	~15% of overlapping partner docs; legal vs display advertiser names	source precedence rules
At-least-once duplicates / malformed / schema evolution	the event stream	stream processing discipline

10. Suggested target (Gold) star schema

Instructor guidance — one reasonable dimensional model students should arrive at (not simply copying the raw structure):

- **Dimensions:** dim_advertiser (SCD2 on account_manager, built from Mongo am_history), dim_partner (CONFORMED across partners_raw + partner_profiles with an explicit precedence rule), dim_offer, dim_campaign, dim_creative, dim_geo, dim_device, dim_date (role-playing: conversion / click / impression / adjusted), and a junk dimension for the flag columns.
- **Facts:** fact_conversion (grain = one de-duplicated conversion — MAX(updated_at); measures: revenue net of adjustments, payout, margin; degenerate conversion_id), fact_click and fact_impression (factless), bridge_conversion_touchpoint for multi-touch.
- **Stretch:** a streaming path — replay the event log incrementally (by offset watermark) into the same fact_conversion, proving batch and stream produce identical Gold.

11. How it maps to the DWDF curriculum

Dataset feature	Course weeks
Normalise the unnormalised relational source	W1 (normalisation UNF→3NF)
Bronze (raw) → Silver (clean) → Gold (star)	W2, W3, W7 (Medallion)
Star schema · grain · surrogate keys	W3, W6
SCD2 from Mongo am history + embedded signal	W3, W6
Additivity + all fact types (transactional / factless / bridge)	W6
Snowflake chain + bridge / M:N (the W6 gaps)	W6
created_at/updated_at watermarks → incremental & idempotent	W7 (+ Mini-Project 02)
Document source: JSONL ingestion, unnesting, schema drift	W7 (modern integration)
Event stream: at-least-once, ordering, replay, schema evolution	W7 (CDC & streaming — hands-on, no infra)
ETL in NiFi / orchestration in Airflow	W4, W7
Data mart + Metabase dashboards on Gold	W8 (OLAP / BI)
Governance, lineage, DQ tests, PII tagging (ip / user_agent / contact)	W9 (OpenMetadata)

12. Loading & package contents

```
PulseTrack_Capstone/
  data/                          # 11 relational CSVs (774,344 rows)
  ddl/pulsetrack_source_schema.sql
  mongo/partner_profiles.jsonl   # 600 docs
  mongo/advertiser_accounts.jsonl# 60 docs
  stream/pulsetrack_conversions.log   # 183,162 lines
  stream/README_topic.md         # topic format & semantics
  answer_key/ground_truth.json   # INSTRUCTOR ONLY — final states, canon maps, totals
  generator/gen.py               # seeded generator (reproducibility, seed 42)
  generator/er_diagram.py        # regenerates the ER figure
  generator/qa.py (+ build tools) # instructor QA suite & doc builders
  pulsetrack_er_diagram.png
  PulseTrack_Data_Guide.docx     # this file
```

Relational load (PostgreSQL):

```
createdb pulsetrack_raw
psql -d pulsetrack_raw -f ddl/pulsetrack_source_schema.sql
# then, in psql, for each table:
\copy pulsetrack_raw.conversions_raw FROM 'data/conversions_raw.csv' WITH (FORMAT csv, HEADER true)
```

Notes for the instructor

- Reproducible: generator/gen.py (fixed seed 42) regenerates ALL THREE sources + the answer key identically; generator/er_diagram.py regenerates the ER figure.
- answer_key/ground_truth.json holds the per-conversion final status / revenue / net revenue, the canonical partner map (with in_rds / in_mongo flags), the AM history, and grading totals — keep it out of the student hand-out.
- PII hooks for the Week-9 governance lab: clicks_raw.ip / user_agent, and contact details inside partner_profiles.

> **Note:** *This package is the DATASET only. The project brief (questions, marks, deliverables) is yours to build around it — matching the ABC Hub Mini-Project format.*